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Natural Language Processing Lab 4

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In this Lab Demo we will cover:

- Word Senses
- Relations between Senses
- WordNet: A Database of Lexical Relations
- Applications of WordNet
- NLTK WordNet Implementation



Word Sense

In [linguistics](#), a **word sense** is one of the meanings of a [word](#).

Words are in two sets:

- Large set with multiple meanings (word senses)
- Small set with only one meaning (word sense).

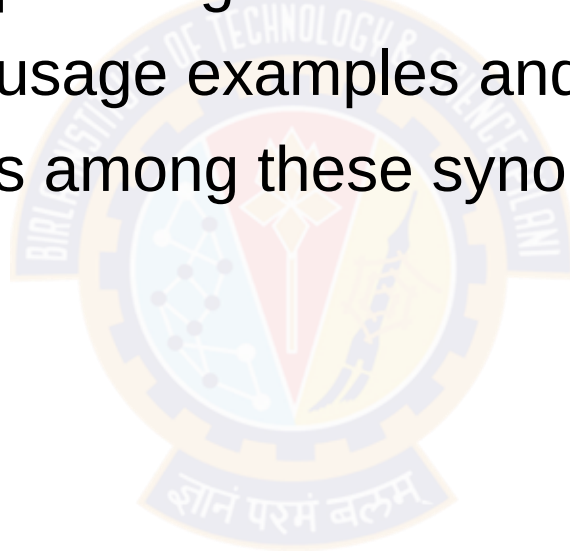
For example, a [dictionary](#) may have over 50 different senses of the word "[play](#)", each of these having a different meaning based on the [context](#):

- We went to see the **play** *Romeo and Juliet* at the theater.
- The coach devised a great **play** that put the visiting team on the defensive.
- The children went out to **play** in the park.

In each sentence we associate a different meaning of the word "play" based on hints the rest of the sentence gives us.

Wordnet

- "WordNet" is a lexical database for the English language.
- Nouns, verbs, adjectives and adverbs are grouped into sets of cognitive synonyms (synsets), each expressing a distinct concept.
- Provide short definitions and usage examples and
- Records a number of relations among these synonym sets or their members.



Synset example

- “car” as in
 - {car, auto, automobile, machine, motorcar}
 - {car, railcar, railway car, railroad car}.
- “Chocolate” as in-

Synset	Gloss
{cocoa, chocolate hot chocolate}	(made from baking chocolate or cocoa powder and milk and sugar)
{chocolate}	(made from roasted ground cacao beans)
{chocolate, coffee, deep brown umber, burnt umber}	(a medium to dark brown color)

Word senses and relationships between word senses

- Polysemy
- Synonymy
- Antonymy
- Hypernymy
- Hyponymy

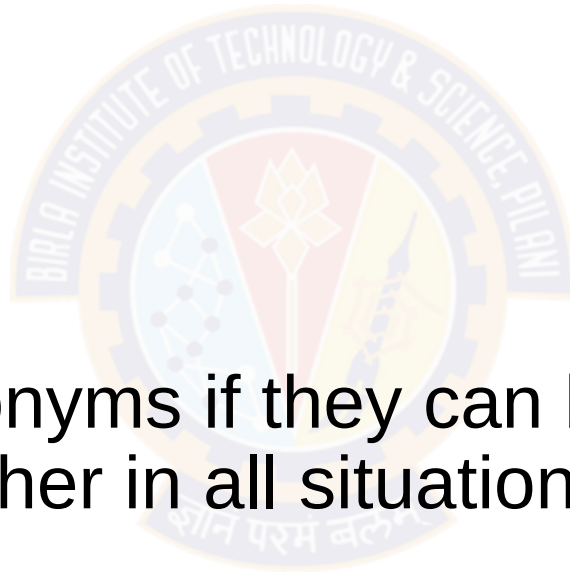


Polysemy

- A single lexeme with multiple meanings
- The **bank** is constructed from red brick
- I withdrew the money from the **bank**
 - Which sense of bank is this?
 - Is it distinct from the river bank sense?
 - How about the savings bank sense?

Synonyms

- Word that have the same meaning in some or all contexts.
 - couch / sofa
 - big / large
 - automobile / car
 - Water / H₂O
- Two lexemes are synonyms if they can be successfully substituted for each other in all situations



Antonyms

- Senses that are opposites with respect to one feature of their meaning
 - dark / light
 - short / long
 - hot / cold
 - up / down
 - in / out
- More formally: antonyms can
 - define a binary opposition or at opposite ends of a scale (*long/short, fast/slow*)
 - Be **reverses**: *rise/fall, up/down*



Hyponym and Hypernym

- One sense is a **hyponym** of another if the first sense is more specific, denoting a subclass of the other
 - *car* is a hyponym of *vehicle*
 - *dog* is a hyponym of *animal*
 - *mango* is a hyponym of *fruit*
- Conversely
 - *vehicle* is a hypernym/superordinate of *car*
 - *animal* is a hypernym of *dog*
 - *fruit* is a hypernym of *mango*

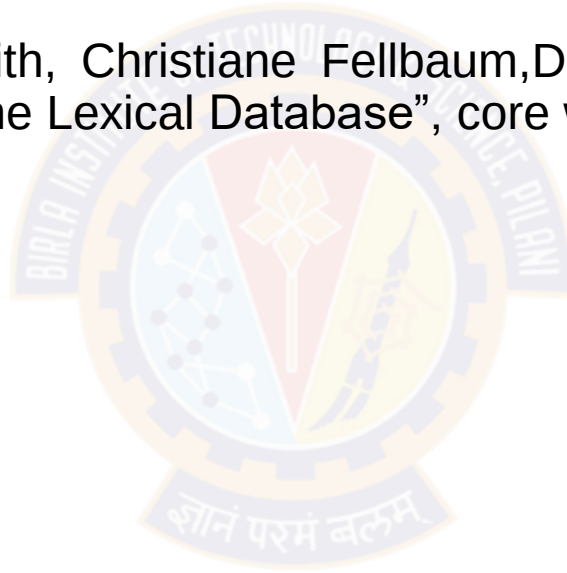
hypernym	vehicle	fruit	furniture	mammal
hyponym	car	mango	chair	dog

Applications of WordNet

- Word-sense disambiguation
 - neighbouring words will be more closely related to correct sense (desert/dessert ~ camel)
- Document classification
 - What is this text about? Look for recurring hypernyms
- Document retrieval
 - looking for texts about sports cars, search for synonyms and hyponyms of *sports car*
- Open-domain Q/A
 - Searching texts (eg WWW) to answer questions expressed in natural language
- Textual entailment
 - Answering questions implied by text

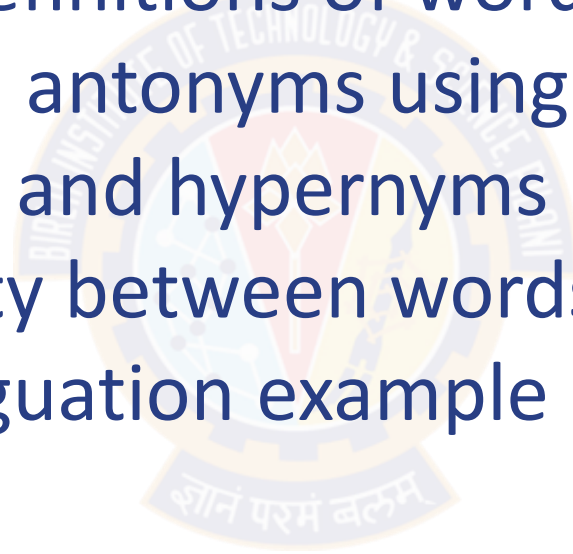
References

1. WordNet: <http://wordnet.princeton.edu/>
2. Christiane Fellbaum, MIT, "WordNet : an electronic lexical database", MIT Press
3. George A. Miller, Richard Beckwith, Christiane Fellbaum, Derek Gross, and Katherine Miller, "Introduction to WordNet: An On-line Lexical Database", core working paper



WordNet Lab Demo

- Implementation of WordNet in NLTK
- Extracting synsets, definitions of words
- Extracting synonyms, antonyms using WordNet
- Extracting hyponyms and hypernyms using WordNet
- Finding path similarity between words in WordNet
- Word sense disambiguation example





Thank you